Introduction to Data Analysis

Learning Objectives:

- Understand the difference between numerical and categorical data.
- Calculate basic summary statistics.
- Identify missing values and possible outliers.
- Create clear charts for a non-technical audience.

Important Definitions:

Dataset: A structured collection of observations and variables.

Numerical variable: A field containing measurable quantities, such as age, income, or loan amount.

Categorical variable: A field representing groups, such as gender, education, or property area.

> Good analysis starts by checking the quality and meaning of the data before calculating results.

Recommended Workflow:

1. Import the dataset.
2. Inspect the rows and columns.
3. Check data types and missing values.
4. Calculate descriptive statistics.
5. Visualize important patterns.
6. Record conclusions and limitations.

Practical Checklist:

Status	Task
Complete	Confirm the file opens correctly.
Complete	Review column names.
Pending	Investigate missing credit-history values.
Pending	Compare approval rates by property area.

Example Formulas:

The arithmetic mean is calculated as the sum of all observations divided by the number of observations.

```
mean = sum(values) / count(values)
approval_rate = approved_applications / total_applications
```

Class Discussion:

Why can an average be misleading when the dataset contains extreme values? Consider whether the median would provide a more representative measure.

Homework:

Prepare a one-page summary containing three findings, one chart, and one data-quality limitation.